

AI Accelerator Design | Assignment 3

Manual BCHW to B x (C*H*W) mapping and reconstruction

1. Objective and address mapping

This work maps a batch of channel-first image or feature-map tensors from shape $B \times C \times H \times W$ into $B \times (C*H*W)$, then reconstructs the original layout using explicit loops and index arithmetic. NumPy provides storage and element access; both conversions are implemented element by element.

For a fixed batch index b , the flattened position is $j = c*H*W + h*W + w$. The inverse is $c = j // (H*W)$, $r = j \% (H*W)$, $h = r // W$, and $w = r \% W$. Batch index b is preserved as the row index of the two-dimensional result.

2. Pseudocode

```
FLATTEN(I, B, C, H, W):
    allocate F[B, C*H*W]
    for b = 0..B-1, c = 0..C-1:
        for h = 0..H-1, w = 0..W-1:
            j = c*H*W + h*W + w
            F[b,j] = I[b,c,h,w]
    return F

RECONSTRUCT(F, B, C, H, W):
    allocate R[B, C, H, W]
    for b = 0..B-1, j = 0..C*H*W-1:
        c = j // (H*W); r = j % (H*W)
        h = r // W; w = r % W
        R[b,c,h,w] = F[b,j]
    return R
```

3. Manual indexing example

Let $B=1$, $C=2$, $H=2$, $W=3$, with channel 0 equal to $[[a,b,c],[d,e,f]]$ and channel 1 equal to $[[g,h,i],[j,k,l]]$. The flattened row is $[a,b,c,d,e,f,g,h,i,j,k,l]$. For the element $I[0,1,1,2]=l$, $j=1*2*3 + 1*3 + 2=11$, so $F[0,11]=l$. In reverse, $11 // 6=1$, remainder 5, $5 // 3=1$, and $5 \% 3=2$, recovering $(c,h,w)=(1,1,2)$.

4. Experiments and results

The grayscale and RGB inputs are 64×96 crops from scikit-learn's bundled china.jpg sample image; grayscale values use the standard $0.299R + 0.587G + 0.114B$ luminance formula. Synthetic feature maps use $B=2$, $H=W=8$ and seeded uniform floating-point values. The seed is 20261005 for reproducibility.

Dataset / input	B	C	H	W	E _{max}	MAE
Sample photo, grayscale crop	1	1	64	96	0.0e+00	0.0e+00
Sample photo, RGB crop	1	3	64	96	0.0e+00	0.0e+00
Synthetic feature map C=8	2	8	8	8	0.0e+00	0.0e+00
Synthetic feature map C=16	2	16	8	8	0.0e+00	0.0e+00
Synthetic feature map C=32	2	32	8	8	0.0e+00	0.0e+00
Synthetic feature map C=64	2	64	8	8	0.0e+00	0.0e+00
Synthetic feature map C=128	2	128	8	8	0.0e+00	0.0e+00
Synthetic feature map C=256	2	256	8	8	0.0e+00	0.0e+00
Synthetic feature map C=500	2	500	8	8	0.0e+00	0.0e+00

5. Discussion

All tested configurations reconstructed exactly: $E_{max}=0$ and $MAE=0$ in every row. This includes the single-channel grayscale case, three-channel RGB case, and feature maps up to 500 channels. The result confirms that the C-H-W ordering and the inverse quotient/remainder calculations agree for all tested indices. The experiments check data-layout correctness; they do not measure accelerator throughput or memory bandwidth.